

Title page

Title

OligoVigil: a curator-verified, source-anchored database of safety and off-target evidence for therapeutic oligonucleotides

Running title (≤50 chars)

OligoVigil: curated ASO/siRNA safety evidence (48 characters)

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Declarations summary

- **Conflict of interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Data availability:** Release data (737 curator-verified records: 626 toxicity + 111 off-target; audit trail, benchmark splits, deterministic baselines), schema, API and code will be openly available without login at [public HTTPS URL TBD]; the versioned release snapshot is archived at Zenodo DOI 10.5281/zenodo.20633779. Full statement: `declarations/03_data_availability.md`.
- **Inter-rater agreement:** A second independent curator reviewed a 100-row mixed accept+reject stratified sample (KAPPA-2); binary Cohen $\kappa = 0.34$ (fair, per Landis-Koch); the second curator is systematically stricter than the curator of record (a safety-conservative direction). The pilot 100-row KAPPA-1 release-only round (κ not computable; Grade-axis $\kappa = 0.21$) is superseded. Full inter-rater treatment in §Methods Stage 3.
- **Code availability:** Curation scripts, QA suites, baseline reproducibility code and the portal source are released under the MIT License (code) and CC BY 4.0 (derived annotations); see LICENSE and LICENSE-DATA in the repository.
- **Ethics:** No human subjects, animal experiments or identifiable personal data were involved; the resource is derived from published preclinical literature.
- **AI-assisted curation disclosure:** A documented AI-assisted human curation workflow was used (Stage-2 source-grounded LLM proposals with verbatim grounding gate, followed by single-human

adjudication); the firewall between machine proposals and human decisions is enforced in the data model and audited per record. See `declarations/06_genai_disclosure.md`.

- **Author contributions (CRediT):** see `declarations/01_credit_author_statement.md`. Summary: **Jie Ni** — Conceptualisation; Methodology; Software; Investigation; Formal analysis; Data curation; Visualisation; Writing — original draft; Project administration. **Xinting Zhang** — Investigation; Data curation; Validation. **Zhuoying Xie** — Investigation; Visualisation. **Shan Lu** — Resources; Writing — review & editing. **Yun Liu** — Conceptualisation; Supervision; Writing — review & editing; Resources; Funding acquisition. **Adam Jatowt** — Supervision; Writing — review & editing; Resources.

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- Abstract: **192 words**
- Main text (Introduction through Future directions, excluding figure captions, declarations and references): **2,987 words**
- Number of figures: **6** (Figure 1: curation pipeline; Figure 2: evidence-object architecture; Figure 3: evidence landscape; Figure 4: web portal walkthrough; Figure 5: peer comparison; Figure 6: validation dashboard)
- Number of tables: **0**
- Number of references: **31** verified references; no citation placeholders in the main manuscript

Submission

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